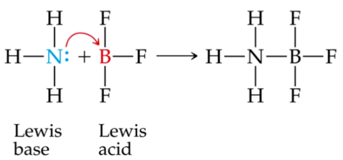
① Three Definitions

Definition	Acid	Base
Arrhenius	H^+ producer in water	OH^- producer in water
Brønsted-Lowry	H^+ donor	H^+ accepter
Lewis	e^- pair acceptor	e^- pair donor

amphiprotic 既是酸 又是碱 (针对B类) ⟶ 水

Lewis Acids and Lewis Bases

- Lewis bases are defined as electron-pair donors.
- Anything that could be a Brønsted–Lowry base is Lewis base.
- Lewis bases can interact with things other than protons, however.
- Atoms with an empty valence orbital can be Lewis acids.



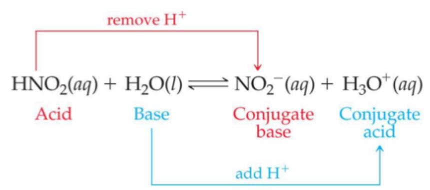
conjugate 共轭

autoionization 自耦电离 水的自耦电离是吸热,

② 共轭酸碱对

- **The stronger an acid, the weaker its conjugate base; the stronger a base, the weaker its conjugate acid.**

- The term **conjugate** 共轭 means "joined together as a pair."
- Reactions between acids and bases always yield their conjugate bases and acids.



③ 计算

At 25°C $K_w = [H_3O^+][OH^-] = 1\times10^{-14}$

其它温度时，kw不是这个值，会随温度变化而变化

- If $[H^+] = [OH^-]$, a solution is neutral.
- If $[H^+] > [OH^-]$, a solution is acidic.
- If $[H^+] < [OH^-]$, a solution is basic.

$pH = -\log[H^+]$

$pOH = -\log[OH^-]$ ★小数点后2位

$pH + pOH = pK_w = 14$ (25°C)

↳ 如何测量 pH

- Acid — Base indicators 酸碱指示剂
- pH meters pH 剂

④ 酸碱强弱

1° 强酸 (7大)

HCl、HBr、HI、HNO_3、H_2SO_4、$HClO_3$、$HClO_4$ (注意HF不是)

$HA + H_2O \rightarrow H_3O^+ + A^-$

2° 强碱

可溶且碱金属、或碱土金属

$NaOH$、KOH、$Ca(OH)_2$、$Sr(OH)_2$、$Ba(OH)_2$

$MOH(aq) \longrightarrow M^+(aq) + OH^-(aq)$

$M(OH)_2(aq) \longrightarrow M^{2+}(aq) + 2OH^-(aq)$

3° 弱酸

2种表达式

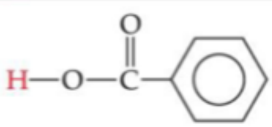
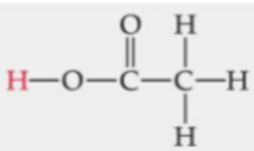
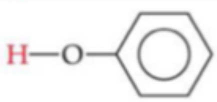
$$HA(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + A^-(aq)$$
or $$HA(aq) \rightleftharpoons H^+(aq) + A^-(aq)$$

- Since it is an equilibrium, there is an equilibrium constant related to it, called the **acid-dissociation constant**, K_a:

$$K_a = [H_3O^+][A^-] / [HA]$$
or $$K_a = [H^+][A^-] / [HA]$$

➢ Weak Acids

Table 16.2 Some Weak Acids in Water at 25 °C

Acid	Structural Formula*	Conjugate Base	K_a
Chlorous ($HClO_2$)	H—O—Cl—O	ClO_2^-	1.0×10^{-2}
Hydrofluoric (HF)	H—F	F^-	6.8×10^{-4}
Nitrous (HNO_2)	H—O—N=O	NO_2^-	4.5×10^{-4}
Benzoic (C_6H_5COOH)		$C_6H_5COO^-$	6.3×10^{-5}
Acetic (CH_3COOH)		CH_3COO^-	1.8×10^{-5}
Hypochlorous (HOCl)	H—O—Cl	OCl^-	3.0×10^{-8}
Hydrocyanic (HCN)	H—C≡N	CN^-	4.9×10^{-10}
Phenol (HOC_6H_5)		$C_6H_5O^-$	1.3×10^{-10}

*The proton that ionizes is shown in red.

Acids and Bases

计算

1° 从pH → Ka

先 → [H] = [O]

$[H^+] = [O]$

看与初始浓度相比是否 < 5% $\left(\frac{1}{20}\right)$

2° → Percent ionization

- Percent ionization = $\frac{[H_3O^+]_{eq}}{[HA]_{initial}} \times 100\%$

3° 从 $K_a \longrightarrow pH$

从 $\frac{x^2}{O - x} = K_a$

$\frac{x}{O} < \frac{1}{20}$ | 5% 则

$\frac{x^2}{O} = K_a$

$-\log_{10}\sqrt{K_a \cdot O}$

否则正常解方程

Polyprotic 多元

如果 $K_{a_1} / K_{a_2} \gg 10^3$

PH的计算只依靠一级溶解

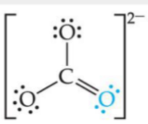
$K_{a_1} > K_{a_2}$

4° 弱碱

$\square_{(aq)} + H_2O_{(l)} \rightleftharpoons O_{(aq)} + OH^-_{(aq)}$

base dissociation constant

Table 16.4 Some Weak Bases in Water at 25 °C

Base	Structural Formula*	Conjugate Acid	K_b
Ammonia (NH_3)	H—N̈—H, with H on N	NH_4^+	1.8×10^{-5}
Pyridine (C_5H_5N)		$C_5H_5NH^+$	1.7×10^{-9}
Hydroxylamine ($HONH_2$)	H—N̈—ÖH, with H on N	$HONH_3^+$	1.1×10^{-8}
Methylamine (CH_3NH_2)	H—N̈—CH_3, with H on N	$CH_3NH_3^+$	4.4×10^{-4}
Hydrosulfide ion (HS^-)	[H—S̈:]$^-$	H_2S	1.8×10^{-7}
Carbonate ion (CO_3^{2-})		HCO_3^-	1.8×10^{-4}
Hypochlorite ion (ClO^-)	[:C̈l—Ö:]$^-$	HClO	3.3×10^{-7}

*The atom that accepts the proton is shown in blue.

ducation

Ka、Kb的关系

For a conjugate acid–base pair, K_a and K_b are related in this way:

$$K_a \text{ (an acid)} \times K_b \text{ (its conjugate base)} = K_w$$

⑤盐的酸碱性

hydrolysis 水解

分开看

阴离子:

1° 强酸的阴离子是酸性 (Cl^-)

2° 弱酸的阴离子是碱性 (CH_3COO^-)

3° 多元酸的阴离子要比较前后Ka、Kb的大小

阳离子:

1° 强碱的阳离子是中性 (Ca^{2+}、Sr^{2+}、Ba^{2+})

2° 弱碱的阳离子是酸性 (NH_4^+)

3° 过渡金属形成阳离子是酸性 ← Transition / Post-transition metal

hydrated cations 水合阳离子

aqua ligand 水合配体

水吸附，酸性更强

Table 16.6 Acid-Dissociation Constants for Metal Cations in Aqueous Solution at 25 °C

Cation	K_a
Fe^{2+}	3.2×10^{-10}
Zn^{2+}	2.5×10^{-10}
Ni^{2+}	2.5×10^{-11}
Fe^{3+}	6.3×10^{-3}
Cr^{3+}	1.6×10^{-4}
Al^{3+}	1.4×10^{-5}

结合起来看盐

如果一方呈中性，整体由另一方决定

如果两方都有酸碱性，则比较Ka、Kb，谁大由谁

⑥影响酸性强弱的因素

①无氧的 (Binary Acids)

H - X

↳ Bond strength 决定

越弱 (键能越小) → 越容易断裂 → 酸越强

对于-族 键能越小，酸性越强，从上往下 H⁺↑，(HI > HBr > HCl > HF)

对于-周期 受 bond polarity 影响，从左往右 H⁺↑

②有氧的 oxyacids 含氧酸

- Generally, as the electronegativity of the nonmetal Y increases, the acidity increases for acids with the same structure.

1° 电负性越大，酸性越强

2° 对同一个□，O越多，酸性越强

electronegativity 电负性 (氧化数越大)

O 本身电负性挺大的

⑦ carboxylic acid 羧酸

有 - COOH

resonance

hydronium ion 水合H⁺ H_3O^+

HS^- 呈碱性